UNIT 3

FRACTIONS

A fraction is a part of a whole. In arithmetic, the number is expressed as a quotient, in which the numerator is divided by the denominator.

ADDITION

Example 1. Add $\frac{2}{5}$ and $\frac{1}{4}$

Solution:

$$\frac{2}{5} + \frac{1}{4}$$

Find LCM of the denominators

2	5, 4
2	5, 2
5	5, 1
	1, 1

$$\frac{2}{5} + \frac{1}{4} = \frac{(2 \times 4) + (1 \times 5)}{20}$$

$$= \frac{8 + 5}{20}$$

$$= \frac{13}{20}$$

LCM = 2 x 2 x 5 = 20

EXERCISE 3.1

Q1. Add the following.

1) $\frac{5}{6} + \frac{4}{5} =$

2) $\frac{5}{8} + \frac{4}{6} =$

3) $\frac{5}{6} + \frac{4}{9} =$

4) $\frac{1}{4} + \frac{2}{7} =$

5) $\frac{7}{7} + \frac{4}{6} =$

6) $\frac{5}{4} + \frac{3}{9} =$

Q2. Solve the following.

1) $\frac{3}{4} + \frac{7}{12} =$

2) $\frac{5}{6} + \frac{1}{3} =$

3) $\frac{6}{7} + \frac{5}{21} =$

4) $\frac{7}{8} + \frac{5}{8} =$

5) $\frac{3}{5} + \frac{11}{20} =$

6) $\frac{7}{10} + \frac{23}{50} =$

Q3. Solve the following.

1) $\frac{7}{9} + \frac{2}{9} + \frac{5}{9} =$

2) $\frac{1}{5} + \frac{2}{5} + \frac{1}{5} =$

3) $\frac{5}{12} + \frac{2}{12} + \frac{6}{12} =$

4) $\frac{3}{10} + \frac{9}{10} + \frac{8}{10} =$

5) $\frac{6}{7} + \frac{6}{7} + \frac{4}{7} =$

6) $\frac{4}{7} + \frac{3}{7} + \frac{5}{7} =$

7) $\frac{6}{9} + \frac{2}{9} + \frac{6}{9} =$

8) $\frac{8}{12} + \frac{6}{12} + \frac{7}{12} =$

SUBTRACTION

Example 1. Subtract $\frac{2}{7}$ from $\frac{1}{3}$

Solution:

$$\frac{1}{3} - \frac{2}{7} = \frac{(1 \times 7) - (2 \times 3)}{21}$$

$$= \frac{7-6}{21}$$

$$= \frac{1}{21}$$

3	3, 7
7	1, 7
	1, 1

LCM = 3 x 7 = 21

Thus, $\frac{1}{3} - \frac{2}{7} = \frac{1}{21}$

EXERCISE 3.2

Q1. Subtract the following.

1) $\frac{6}{7} - \frac{5}{7} =$

2) $\frac{4}{6} - \frac{2}{6} =$

3) $\frac{4}{6} - \frac{1}{6} =$

4) $\frac{3}{3} - \frac{1}{3} =$

5) $\frac{4}{7} - \frac{1}{7} =$

6) $\frac{6}{6} - \frac{1}{6} =$

7) $\frac{2}{3} - \frac{1}{3} =$

8) $\frac{2}{2} - \frac{1}{2} =$

Q2. Subtract the following.

1) $\frac{23}{24} - \frac{1}{4} - \frac{2}{12} =$

2) $\frac{53}{54} - \frac{4}{6} - \frac{1}{27} =$

3) $\frac{30}{32} - \frac{2}{4} - \frac{1}{16} =$

4) $\frac{23}{24} - \frac{1}{3} - \frac{2}{12} =$

5) $\frac{16}{18} - \frac{4}{6} - \frac{1}{9} =$

6) $\frac{43}{45} - \frac{3}{5} - \frac{1}{15} =$

Q1. Solve the following.

1) $\frac{2}{3} + \frac{1}{2} =$

2) $\frac{3}{5} - \frac{3}{10} =$

3) $\frac{2}{5} + \frac{1}{10} =$

4) $\frac{4}{6} - \frac{1}{12} =$

5) $\frac{1}{5} + \frac{2}{4} =$

6) $\frac{4}{6} - \frac{1}{2} =$

MULTIPLICATION

Example 1. Multiply $\frac{1}{2}$ by $\frac{4}{5}$

Solution:

So, $\frac{1}{2} \times \frac{4}{5}$

$$= \frac{1 \times \cancel{4}^{2}}{\cancel{2}_{1} \times 5} = \frac{2}{5}$$

Example 2. Simplify.

1) $2\frac{2}{3} \times \frac{9}{4}$

Solution:

$$2\frac{2}{3} \times \frac{9}{4} = \frac{8}{3} \times \frac{9}{4}$$

$$= \frac{{}^{2}\cancel{8} \times \cancel{9}^{3}}{{}_{1}\cancel{3} \times \cancel{4}_{1}}$$

$$= \frac{2 \times 3}{1 \times 1}$$

$$= \frac{6}{1}$$

$$= 6$$

Example 3.

2) $\frac{6}{5} \times \frac{25}{8} \times \frac{1}{3}$

Solution:

$$\frac{6}{5} \times \frac{25}{8} \times \frac{1}{3}$$

$$= \frac{{}^{2}\cancel{6}^{1} \times \cancel{25}^{5} \times 1}{{}_{1}\cancel{5} \times \cancel{8}_{4} \times \cancel{3}_{1}}$$

$$= \frac{1 \times 5 \times 1}{1 \times 4 \times 1}$$

$$= \frac{5}{4}$$

$$= 1\frac{1}{4}$$

EXERCISE 3.3

Q1. Multiply the fraction by the given whole number.

1) $\frac{3}{4} \times 4$	**2)** $\frac{1}{4} \times 16$	**3)** $\frac{3}{5} \times 5$
4) $\frac{1}{2} \times 4$	**5)** $\frac{1}{4} \times 8$	**6)** $\frac{3}{2} \times 4$
7) $\frac{1}{3} \times 6$	**8)** $\frac{2}{3} \times 6$	**9)** $\frac{3}{5} \times 25$

Q2. Multiply the given fraction by another fraction.

1) $\frac{1}{2} \times \frac{2}{4}$

2) $\frac{2}{6} \times \frac{1}{6}$

3) $\frac{4}{5} \times \frac{1}{7}$

4) $\frac{2}{3} \times \frac{2}{5}$

5) $\frac{2}{3} \times \frac{1}{2}$

6) $\frac{2}{8} \times \frac{2}{4}$

7) $\frac{1}{3} \times \frac{1}{3}$

8) $\frac{3}{6} \times \frac{4}{6}$

Q3. Multiply the following.

1) $\frac{3}{5} \times \frac{1}{3}$

2) $\frac{2}{5} \times \frac{5}{10}$

3) $\frac{2}{5} \times \frac{9}{10}$

4) $\frac{2}{4} \times \frac{2}{3}$

5) $\frac{4}{5} \times \frac{7}{10}$

6) $\frac{2}{3} \times \frac{4}{5}$

Q4. Multiply the following.

1) $\frac{8}{5} \times \frac{10}{9} \times \frac{26}{7} =$ ________________

2) $\frac{19}{7} \times \frac{1}{3} \times \frac{19}{7} =$ ________________

3) $\frac{1}{5} \times \frac{5}{2} \times \frac{17}{6} =$ ______________

4) $\frac{1}{2} \times \frac{32}{9} \times \frac{13}{6} =$ ______________

5) $\frac{16}{9} \times \frac{4}{7} \times \frac{1}{5} =$ ______________

6) $\frac{1}{2} \times \frac{5}{3} \times \frac{7}{3} =$ ______________

7) $\frac{11}{6} \times \frac{1}{9} \times \frac{3}{4} =$ ______________

8) $\frac{16}{5} \times \frac{11}{3} \times \frac{8}{9} =$ ______________

DIVISION

Example 1. Simplify $\frac{1}{2} \div 3$

Solution:

$$\frac{1}{5} \div 3 = \frac{1}{5} \div \frac{3}{1}, \quad \frac{1}{5} \div \frac{3}{1} \text{ means } \frac{1}{5} \times \frac{1}{3}$$

$$= \frac{1}{5} \times \frac{1}{3} \quad \left(\tfrac{1}{3} \text{ is reciprocal of } 3\right)$$

$$= \frac{1 \times 1}{5 \times 3} = \frac{1}{15}$$

Example 2. Divide $\frac{9}{16}$ by $\frac{3}{4}$

Solution:

$$\frac{9}{16} \div \frac{3}{4} = \frac{9}{16} \times \frac{4}{3} \quad \left(\frac{4}{3} \text{ is reciprocal of } \frac{3}{4}\right)$$

$$= \frac{9 \times 4}{16 \times 3} = \frac{\overset{3}{\cancel{9}} \times \overset{1}{\cancel{4}}}{\underset{4}{\cancel{16}} \times \underset{1}{\cancel{3}}}$$

$$= \frac{3 \times 1}{4 \times 1} = \frac{3}{4}$$

Example 3. Divide $\frac{7}{4}$ by $\frac{5}{4}$

Solution:

$$\frac{7}{4} \div \frac{5}{4} = \frac{7}{4} \times \frac{4}{5}$$

$$= \frac{7 \times 4}{4 \times 5} = \frac{7 \times \overset{1}{\cancel{4}}}{\underset{1}{\cancel{4}} \times 5}$$

$$= \frac{7}{5} = 1\frac{2}{5}$$

EXERCISE 3.4

Q1. Divide the following.

1) $\frac{1}{5} \div 6$

2) $\frac{2}{7} \div 3$

3) $\frac{4}{9} \div 8$

4) $\frac{3}{4} \div 6$

5) $\frac{5}{9} \div \frac{1}{4}$

6) $\frac{4}{7} \div \frac{2}{5}$

Q2. Solve the following.

1) $\frac{1}{9} \times \frac{3}{2} \times \frac{3}{2}$

2) $\frac{9}{7} \div \frac{3}{2} \times \frac{19}{18}$

3) $\frac{17}{4} \times \frac{9}{4} \div \frac{1}{9}$

4) $\left(\frac{5}{9} \div \frac{5}{17}\right) \div \frac{1}{15}$

5) $\left(\frac{17}{4} \div \frac{9}{4}\right) \div \frac{1}{9}$

6) $\frac{7}{6} \div \left(\frac{1}{2} \div \frac{13}{11}\right)$

7) $\frac{18}{11} \div \frac{9}{14} \times \frac{3}{4}$

8) $\frac{17}{2} \div \frac{7}{6} \times \frac{2}{3}$

WORD PROBLEMS

1. The height of a door is $2\frac{5}{4}$ metres. Out of this, $\frac{1}{8}$ of the part was trimmed off. How much was trimmed off?

2. On Monday, one-tenth of the total number of students were on leave. If the total number of the students are enrolled is 50, how many students were present?

3. A surgeon advised an operating room technician to arrange 12 drips of dextrose solution. The hospital has only $\frac{5}{4}$ of the required dextrose. How many drips are available in the hospital?

4. A piece of wire is $8\frac{1}{3}$ metre long. Find the total length of wire if there are $13\frac{1}{5}$ such small pieces.

5. The cost of 1 kg tomatoes is Rs 45. Find the cost of $13\frac{5}{4}$ kg of tomatoes.

FRACTION USING BODMAS RULE

Preference of solving brackets is as follows:

(i) ( ) parentheses
(ii) { } curly brackets
(iii) [ ] square brackets.

Order of operation using BODMAS rule as:

(i) Division
(ii) Multiplication
(iii) Addition
(iv) Subtraction

Example:

Simplify by using BODMAS rule:

$$\frac{5}{3} \times \left(1\frac{1}{3} - \frac{1}{2}\right) \div \frac{5}{2}$$

Solution:

$$\frac{5}{3} \times \left(1\frac{1}{3} - \frac{1}{2}\right) \div \frac{5}{2} = \frac{5}{3} \times \left(\frac{4}{3} - \frac{1}{2}\right) \div \frac{5}{2}$$

$$= \frac{5}{3} \times \left(\frac{8-3}{6}\right) \div \frac{5}{2} \quad \text{(By taking LCM)}$$

$$= \frac{5}{3} \times \frac{5}{6} \div \frac{5}{2}$$

$$= \frac{5}{3} \times \frac{5}{6} \times \frac{2}{5} \quad \left(\frac{2}{5} \text{ is reciprocal of } \frac{5}{2}\right)$$

$$= \frac{5 \times \cancel{5}^{1} \times \cancel{2}^{1}}{3 \times \cancel{6}_{3} \times \cancel{5}_{1}} = \frac{5 \times 1 \times 1}{3 \times 3 \times 1} = \frac{5}{9}$$

EXERCISE 3.5

Q1. Simplify the following using BODMAS rule:

1) $\frac{1}{5} + \frac{2}{9} \times 4\frac{1}{3} \div 3\frac{1}{4}$

2) $\frac{1}{4} \times \left(\frac{8}{3} + \frac{2}{7}\right)$

3) $\frac{1}{2} + \left(\frac{3}{4} \times 1\frac{7}{33} - \frac{1}{3}\right)$

4) $\left(3\frac{1}{6} - 1\frac{1}{4}\right) \times 2$

5) $\left(\frac{4}{5} - \frac{3}{10}\right) \times \left(\frac{1}{2} + \frac{3}{4}\right)$

6) $\left(2\frac{1}{2} \div \frac{3}{4}\right) \times \frac{3}{7} - \left(\frac{1}{4} + \frac{1}{8}\right)$

7) $1\frac{3}{5} \times \left(\frac{4}{3} - \frac{3}{4} - 2\frac{1}{3}\right) \div 1\frac{2}{7}$

8) $1\frac{1}{6} + \left(2\frac{3}{4} \times 3\frac{1}{3} \div 2\frac{1}{4}\right) - 7\frac{1}{2}$

Q1. Simplify the following using BODMAS rule:

1) $\left(\left(\frac{3}{4} - \frac{3}{8}\right) \times \frac{7}{9}\right) \div \left(\frac{5}{6} + \frac{1}{3} + \frac{1}{2}\right)$

2) $\left(\left(\frac{2}{9} + \frac{4}{9}\right) \div \frac{1}{6}\right) \times \frac{7}{8} - \frac{1}{5} \times \frac{5}{6}$

3) $\left(\frac{5}{9} + \frac{4}{5} - \frac{2}{5}\right) \div \left(\left(\frac{1}{2} \times \frac{4}{9}\right) \div \frac{1}{9}\right)$

4) $\left(\frac{2}{3} + \frac{2}{9}\right) \div \frac{5}{8} - \frac{5}{9} \times \left(\frac{5}{6} + \frac{1}{6}\right)$